Materials Today: Proceedings 38 (2021) 253–258

Contents lists available at ScienceDirect

Materials Today: Proceedings

journal homepage: www.elsevier.com/locate/matpr

Cooling techniques of the PV module: A review

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ARTICLE INFO

Article history:
Received 20 June 2020
Received in revised form 5 July 2020
Accepted 8 July 2020
Available online 13 August 2020

Keywords:
Cooling methods
Module temperature
PV module
Energy
Radiation

ABSTRACT

The electrical efficiency of a solar photovoltaic panel, is low. Sunlight is the essential input to generate electricity, but they put an adverse effect on its performance due to the rise in temperature at the backside of the panel. The rise in temperature is directly related to the sunlight whose energy is converted into heat and may put a negative effect on the electrical parameters and the life span of the cell. The implementation of cooling techniques will result in reducing the excess heat generated and to reduce overall cell temperature. Most of the research has two paths, one is to enhance the efficiency of the solar PV cell and another to ensure a longer life span at the same time. This paper explains the cooling methods i.e. heat sink, heat pipe, PCM material based, microchannel, thermoelectric (Peltier). Different feature and capability of cooling techniques are explained, to provide valuable information and direction for the researcher for future research, to enhance or upgrade the cooling techniques. The finding of review reveals that water-based microchannel gave efficient cooling besides hot water for various application and combination with PCM make it more efficient in both electrical and thermal aspects.
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1. Introduction

There are various sources of renewable energy in our country like solar, biomass, small hydro power [1,2]. Among the various renewable energy sources, the photovoltaic (PV) technology that convert the direct sunlight radiation to electricity has been gaining great popularity, especially in countries with high solar irradiation. Over the past years, PV has shown rapid development in research to make it more efficient and cost-effective in various countries all across the world. This has contributed to the rapid decrease in fossil fuel dependency, which is the primary source of electricity production [1]. Sun radiation is the primary source of all types of renewable energy sources. Mohanty et al. [3] designed three sides roughed solar air heater to capture solar energy. It can convert directly or indirectly in the form of electricity either through photovoltaic (PV) or solar thermal collectors. The solar thermal efficiency ranges from 40% to 60% but while PV ranges from 10 −20%. Silicon solar cell only uses visible range wavelength ranges from 300 nm to 1100 nm (nanometres) to generate electricity [4,5]. Outside the visible wavelength range, the undesired radiated energy from the sun is converted into heat, causing the solar cell temperature to increase.

The electrical power from the solar cell increase with a decrease in the operating temperature. Besides, with it, the lifetime of the PV cell also increases, and the total amount of power generated is also improved. PV cell efficiency mainly depends on the bandgap of the material and the wavelength of the sunlight [6]. One way to improve efficiency is to find the material with a wide bandgap that can harness a wide range of wavelengths. Hardly 5–20% of sun rays entering the solar cell are harnessed into electricity and the rest of the rays are transmitted to the backside of the cell and increase its temperature up to 70 °C.

2. Effect of ambient temperature on module efficiency

The V_{oc} decrease drastically with an increase in the ambient temperature of the solar PV cell above 25 °C and the I_{sc} increases slightly as shown in Fig. 1, as due to the exponential dependence component of the saturation current I_{sat} on the temperature so that I_{sc} increase slightly [6]. The temperature of the solar panel has a direct effect on its ability to generate electricity, this impact is governed through the temperature coefficient. The percentage of temperature coefficient indicates the rise in output with the rise and fall in the temperature against the normal operating conditions

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https://doi.org/10.1016/j.matpr.2020.07.130
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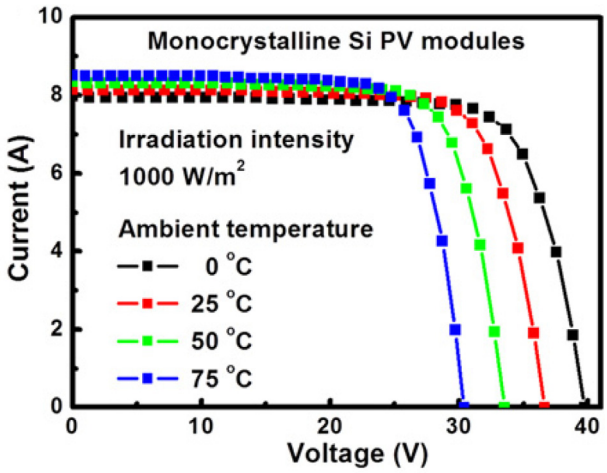


Fig. 1. Current-voltage characteristics at 1000 W/m^2 irradiance [6]

Table 1
Power temperature coefficient of different PV cell technologies.

Technology	P_{mpp} (% / K) MPP power temperature coefficient
Mono- C- Si	−0.40
Multi - C- Si	−0.45
a-Si	−0.26
CIGS	−0.36
CdTe	−0.25

of 25 °C [5]. Table 1 describes some of the commercial PV technology with their Maximum Power Point (MPP) and their temperature coefficients. Negative power coefficient means with every rise of temperature there is a decrease in power output.

Due to all this reason, there is a requirement to remove the heat from the PV, using appropriate cooling methods to solve the negative effect of cell temperature and control the working temperature.

3. Heat extraction methods in the solar PV module

3.1. Need of cooling

The change in surface temperature is influence by various environmental parameters such as sunlight intensity, dust accumulation on its surface, wind direction, humidity. The PV panel is exposed to an open environment to harness more power and all these factors are uncontrollable in an open environment, as there is an optimal solution to control the temperature by external means. To make the panel more efficient, by limiting the temperature rise, cooling methods are applied, which must be more efficient, reliable, economical for a large application.

3.2. Heat sink/ fins / extended surfaces

A Heat sink is an object or device to absorbs and transfer the heat from an object using thermal contact (either direct or radiant). The heat is dissipating by ambient air, induced draft air column, or by coolant. Heat sinks are made of a good thermal conducting such as copper and aluminium alloy. The heat sink surface must be smooth and flat to minimize the thermal resistance with the object that needs to be cooled. Transverse ribs, V-shaped ribs, arc-shaped roughness are used to enhance the heat transfer coefficient, as the cooling speed could be high [7,8]. Grubišić-Čabo et al. [9] use aluminium fins of (L) shape and paste it on the backside of the PV panel with the help of the thermal conducting paste as shown in

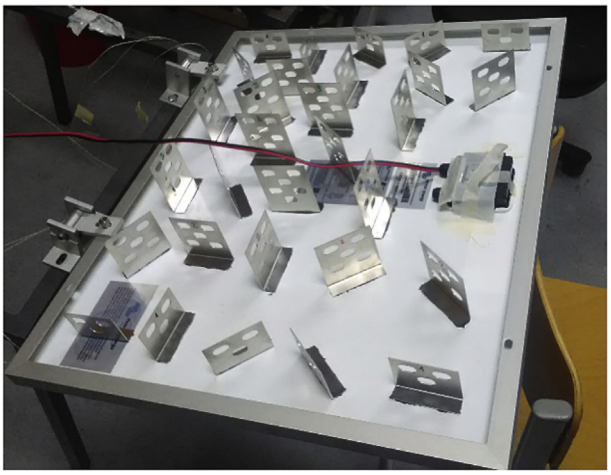


Fig. 2. Backside of PV panel [9]

Fig. 2. During the analysis, it was found that randomly placed fins with holes at the backside give the best cooling of the PV panel as air could penetrate the inner part of the structure with natural blowing air of 1 m/s. Firoozzadeh, Shiravi, and Shafiee [10] done the economic analysis of his experiment for industry and summarized as for 10 KW PV plants, there is a 1.328 KW of power difference which is about 20.22% difference for with and without fins. Table 2 describes the summary of heat sink technology.

3.3. Heat pipe technology

A heat pipe is comprising of three parts an evaporator, an adiabatic transfer section, and a condenser with working fluid inside in both the liquid phase and the gas phase by sharing a common boundary between two things. Behura and Gupta [12] use nanomaterial base fluid for better absorption and heat transfer in solar collectors. Their high thermal conductivity, ease them to bend and shape, and long life makes heat pipe an ideal upgrade to solid heat sink or in some case, an alternative to the pumped liquid cooling system shown in Fig. 3. A heat pipe is a sealed pipe from both sides, made up of a material such as copper, or aluminium with water and ammonia as a working fluid.

Alizadeh et al. [13] use pulsating heat pipes(PHP) as they are compact with high thermal efficiency because of the high amplitude oscillation in gas bubbles and liquid column penetrates in both the condenser and evaporator. Thus, heat transfers by latent heat transfer like ordinary heat pipes and by wiping off the hot walls by the colder moving liquid. PHP inner diameter is 4 mm and covered with polyvinyl fluoride (PVF) layer insulation from the bottom as shown in Fig. 4. A finite-difference method for the solution of the transient heat equation is conducted at a heat flux of 1000 W/m^2 at an immediate surrounding temp of 291 K on monocrystalline silicon solar cells.

Kianifard, Zamen, and Nejad [14] modified a mathematical model and validate it with the experimental data performed in the laboratory. A half-pipe is used to reduce the thermal resistance of a pipe. The result showed that an increase in rows of pipe in serpentine arrangement makes it thermally more efficient and decrease the electrical performance. By increasing the wind speed by 1 to 7 m/s the overall efficiency increased about 3.5%. Table 3 describes some of the latest research in recent years wise.

3.4. Phase change material

Phase change material(PCM) is termed as a latent heat storage capacity of absorbing/delivering the thermal energy when they

Table 2
Summary of the heat sink cooling technologies.

Type of heat extraction	Heat transfer medium	Electrical performance	Thermal performance	Major outcomes	References
Heat sink	Aluminium fins with holes, air-cooled	2% increase due to random placed fins	–	Efficient for low & high solar isolation	[9]
Heat sink	Aluminium fins, air-cooled	2.72% increase in efficiency	Reduction of 7.4 °C from 85 °C	Both economically and experimentally analysis	[10]
Heat sink	Perforated aluminium fins, air-cooled	Increased (V_{OC}) and (P_{MPP}) by 10% and 18.67%	Decrease of temperature from 85.3 °C to 72.8 °C	CFD analysis was done with and without fins	[11]

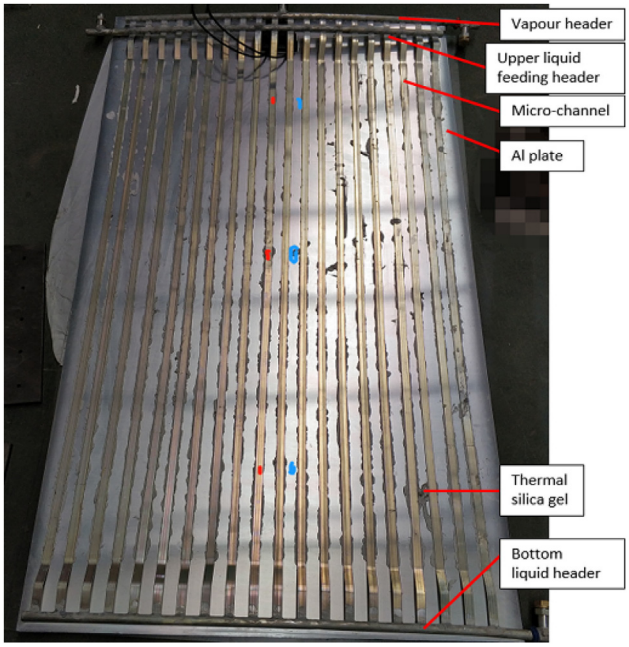


Fig. 3. Heat pipe integration with solar PV panel [15]

undergo the phase transition of state – solid to liquid and vice versa, solid to gas and liquid to gas. Commonly used PCM are organic, inorganic hydrates, various kinds of paraffin, etc. When PCM solidifies it releases energy in terms of latent heat at relatively constant temperature and such material absorbs heat from the environment when they melt. The specific heat capacity of PCM is 50 and it is 12.5 more times water with a value of 4.2 [16]. Its working temp varies in a very wide range of −5 to 190 °C. As salt based PCM may react with the metals and cause corrosion and degradation of the material, and paraffin wax may react with the plastic, melting temperature and melting enthalpy are the two criteria for the best-suited selection of PCM as shown in Fig. 5. PCM chosen for the heat extraction form solar PV must have the melting temperature above the ambient temp to avoid the melting before the rise in temperature at the backside. Due to all these properties, PCM is a better option for the heat extraction from the PV panel as they absorb temp during day time and release that enthalpy during night time as shown in Fig. 6. PCM mixed with nanomaterials gives a better result in terms of thermal and electrical efficiency at very low cost with very high energy storing capacity [17]. Table 4 describes the summary of PCM cooling technology.

3.5. Microchannel heat transfer

Microchannel heat transfer is the heat exchanger in which fluid flows in the lateral direction to the passage with a hydraulic diameter less than 1 mm as shown in Fig. 7. Microchannel is made up of metal or ceramic and the fluid used for transferring the heat from one surface to another is air, water nanofluid, or coolants. Computational fluid dynamics (CFD) analysis is done to find out the optimum flow rate of fluid to achieve the maximum conversion efficiency. Microchannel can also be used as a PVT (Photovoltaic Thermal) in which we store the heat recovered from the panel and used it for other applications. On scaling up the size of the experimental setup and the microchannel the major challenge is to design and manufacture the microchannel for such a large scale. Wang, Wang, and Yuan [24] use microchannel with V-ribs to increase conversion efficiency, heat transfer area, induced chaotic convection, eventually increase the overall heat transfer of the system as shown in Fig. 8. Table 5 describes the summary of microchannel cooling technology.

3.6. Thermoelectric cooling

Thermoelectric cooling is another device that is used to cool down the solar PV panel. As it works on the principle of Peltier effect as based on the seeback effect. They are made up of by joining two types of semiconductor p-type and n-type which act as a

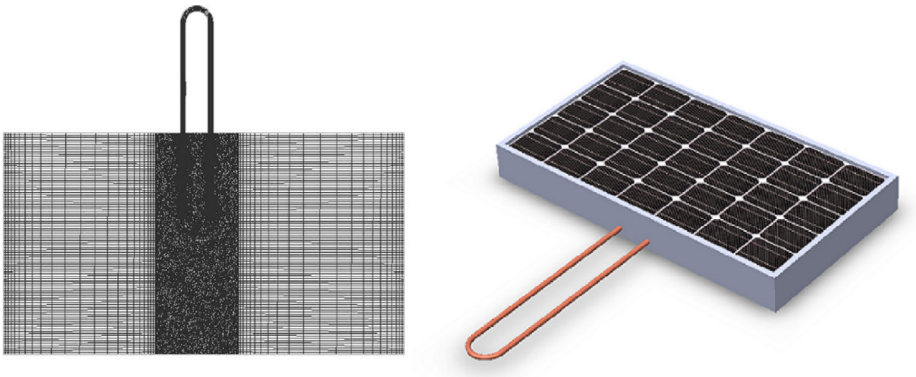


Fig. 4. Model of PV panel with a heat pipe [13]

Table 3
Summary of the heat pipe cooling technologies.

Type of heat extraction	Electrical performance	Thermal performance	Major Outcomes	References
Single turn pulsating heat pipes	Electrical power generated up to 18%	Reduce temperature by 16.1 K	Increase, the efficiency in a shorter time	[13]
Serpentine arrangement half pipe	Increase about 0.4 to 0.6%	Increase of 10 to 13%	Precise mathematical equation is validated with experimental data	[14]
Microchannel loop heat pipe	Increase efficiency from 10% to 16%	Maximum efficiency of 71.67%	Improved evaporation and condensation	[15]

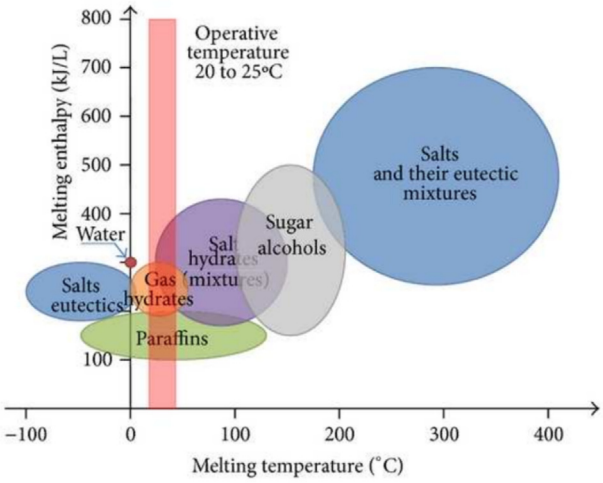


Fig. 5. Graph showing the various PCM with different properties [18]

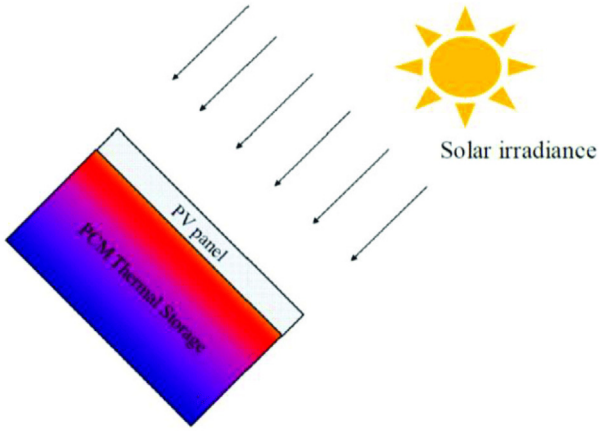


Fig. 6. Cross-sectional view of PCM integrated with panel [19]

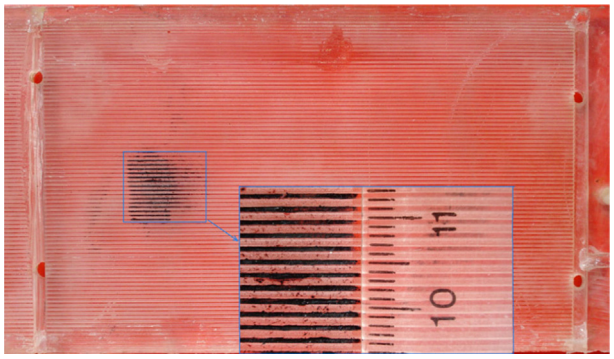


Fig. 7. Microchannel porotype actual picture [25]

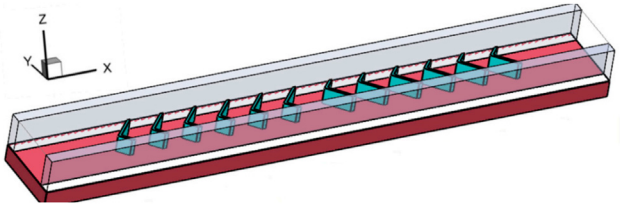


Fig. 8. Microchannel with V – ribs [24]

heat pump which transfers heat from one side to the other side by consuming some of the current in a particular direction. It causes heating on one side and cooling on another side which is being utilized for the temperature reduction of the solar PV panel as shown in Fig. 9.

As they operate on DC therefore we are providing the current from an additional PV panel of particular rating to operate the thermoelectric module (TEM) and heat sink with large fins are connected on the hot side to dissipate the generated heat. As described by [28] in his experiment, the total cost of TEM and accessories associated with is only 6% of the total cost for installing 4 KW, 24 V Off-grid PV system in 2016 in US$, which helps in

Table 4
Summary of PCM cooling technologies.

Type of PCM	Electrical performance	Thermal performance	Major Outcomes	References
RT_{55}	Increase in peak power of 6.7%	13 °C reduction in terms of building-integrated PV module	–	[17]
RT_{55} + 2% Al_2O_3 nanoparticles mixture	Increase in peak power of 8.5%	16 °C reduction in terms of building-integrated PV module	Higher energy storying capacity with lesser cost	[17]
RT28HC	Output power increased by 4.3–8.7%	Lower the temperature by 35.6 °C	Experimental result is validated by simulation on TRNSYS software	[20]
OM-29	Electrical efficiency increased by 2%	Average temperature reduction of 10.35 °C	The experiment performed under direct sunlight for real-time application	[21]
Paraffin wax + heat exchanger	Electrical efficiency increased by 5.18%	Temperature difference reach to 23 °C	Heat extracted can be used for other application	[22]
OM-47 + heat exchanger	Electrical efficiency increased by 14.5%	Reduction in temperature of 33.8%	Thermal efficiency is changing due to gravitational time delay effect and electrical efficiency is same	[23]
OM-47 + heat exchanger + aluminium matrix	Electrical efficiency increased by 15.2%	Reduction in temperature of 40%		[23]

Table 5
Summary of the microchannel cooling technologies.

Type of heat extraction	Heat transfer medium	Electrical performance	Thermal performance	Major outcomes	References
Microchannel heat sink with V-ribs	Nanofluid	–	Transfer capacity increased by 124–202% with ribs in compare of without ribs	Mathematically solved with chaotic convection	[24]
Plexiglas microchannel	Air and water	Maximum output power increased by 38%	Maximum difference of 20 °C	Performance is highly related to slug flow	[25]
Microchannel plate	Water	Maximum output power increased by 30%	Maximum difference of 18.9 °C	High energy performance can be achieved at high Reynolds number	[26]
Micro-channel	Water	Output power increased by 14%	Temperature drop by 15 °C	Validated by CFD and experiment	[27]

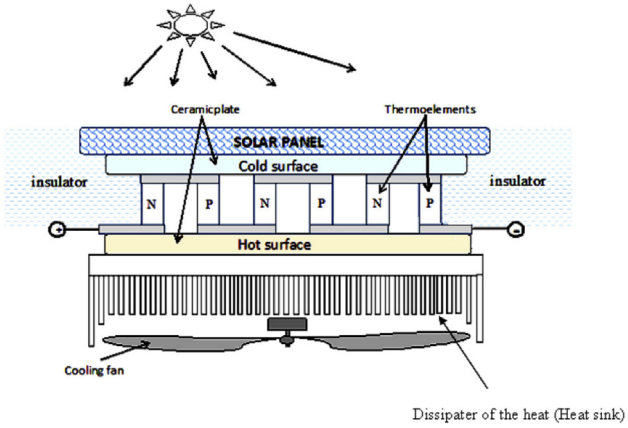


Fig. 9. Constructional view of TEM with PV panel [28]

decreasing the temperature by 18 °C form 83 °C and enhance the life of the solar cell.

Najafi and Woodbury [29] use the genetic algorithm using MATLAB to find out the optimal current for TEM to get maximum output power generated by the system as in his experiment, the TEM is consuming the fraction of current form the main system. He found in his experiment that TEM with a high figure of merit was best suited for the higher solar irradiance.

Borkar, Prayagi, and Gotmare [30] validated the experiment with the CFD analysis and found, the minimum and maximum electrical conversion efficiency of PV panel without TEM is 8.35 and 11.48%, and with TEM is 12.26 and 13.27% respectively. By comparison, he found similar results shown in Fig. 10. He also used a microcontroller for switch ON and OFF process of TEM at 25 °C boundary temperature of the panel.

4. Conclusion

This paper represents an overview of recent studies of solar PV cooling techniques. With an expectation to gain suitable design aspects and future advancements in cooling techniques of PV system for large solar plants. The study reveals that one cooling method is not enough sufficient to establish for large solar PV plants due to the cost involvement and efficiency increased. PCM cooling is best in storing energy in a given region per unit volume. Water-based microchannel gives good cooling with energy recovery as hot water for applications like hospitals, restaurants, and manufacturing industries. As if we combine both PCM and microchannel the heat transfer will increase and overall efficiency will also increase. Thermoelectric cooling can be utilized for remote location plants as there is a requirement of less maintenance and a high life span of the equipment. Future research work should be focused on the economic analysis and test the feasibility with the integration of two cooling methods to find out the best alternative among all.

CRediT authorship contribution statement

Shrey Verma: Writing - original draft, Reviewing. **Subhashree Mohapatra:** Literature review. **Subhankar Chowdhury:** Literature review. **Gaurav Dwivedi:** Conceptualization, Final drafting.

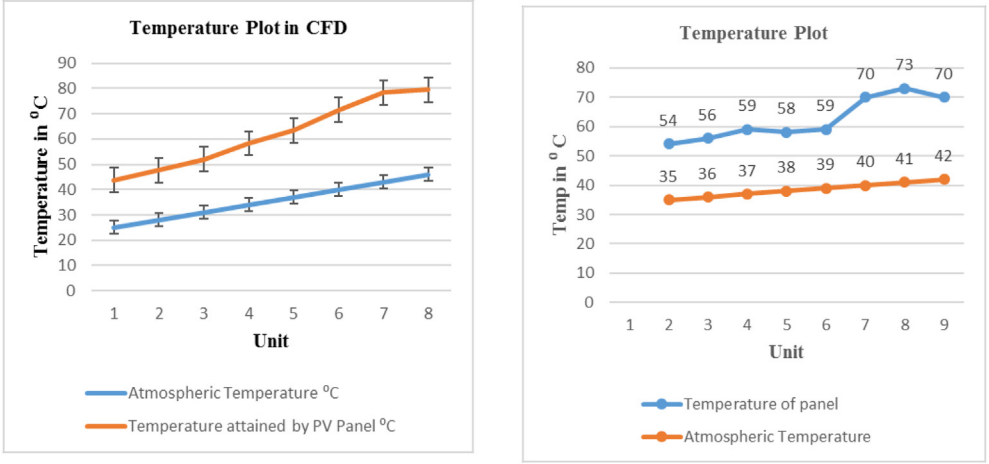


Fig. 10. Comparison of CFD and experimental data [30]

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

I would like to thank MANIT Bhopal and Energy Centre Faculty for supporting the review work.

References

[1] P. Dwivedi, K. Sudhakar, A. Soni, and E. Solomin, " Case Stud. Therm. Eng., vol. 21, no. June, p. 100674, 2020, doi: 10.1016/j.csite.2020.100674.
[2] P. Verma G. Dwivedi A.K. Behura D.K. Patel T.N. Verma A. Pugazhendhi 2020 117920 10.1016/j.fuel.2020.117920.
[3] C. P. Mohanty et al., Energy Sources, Part A Recover. Util. Environ. Eff., no. April, 2020, doi: 10.1080/15567036.2020.1752855.
[4] Y.P. Zhou M.J. Li Y.H. Hu T. Ma "Appl, Energy 260 September 2020 2019, p. 114258 10.1016/j.apenergy.2019.114258.
[5] G. Makrides, Ed. Rijeka: IntechOpen, 2012, p. Ch. 9.
[6] J. A. Jiang, J. C. Wang, K. C. Kuo, Y. L. Su, J. C. Shieh, and J. J. Chou, "Energy, vol. 44, no. 1, pp. 292–301, 2012, doi: 10.1016/j.energy.2012.06.029.
[7] A.K. Behura, B.N. Prasad, L. Prasad, Sol. Energy 130 (2016) 46–59, https://doi.org/10.1016/j.solener.2016.02.006.
[8] B.N. Prasad, A.K. Behura, L. Prasad, Sol. Energy 105 (2014) 27–35, https://doi.org/10.1016/j.solener.2014.03.027.
[9] F. Grubišić-Čabo, S. Nižetić, D. Čoko, I. Marinić Kragić, and A. Papadopoulos, " J. Clean. Prod., vol. 176, pp. 119–129, 2018, doi: 10.1016/j.jclepro.2017.12.149.
[10] M. Firoozzadeh, A. Shiravi, M. Shafiee, Iran J. Energy Environ. 10 (2) (2019) 80–84, https://doi.org/10.5829/ijee.2019.10.02.02.
[11] Z. Arifin, D.D.D.P. Tjahjana, S. Hadi, R.A. Rachmanto, G. Setyohandoko, B. Sutanto, Int J. Photoenergy 2020 (2020), https://doi.org/10.1155/2020/1574274.
[12] A.K. Behura, H.K. Gupta, Mater Today Proc. 22 (2019) 1664–1668, https://doi.org/10.1016/j.matpr.2020.02.183.
[13] H. Alizadeh, R. Ghasempour, M.B. Shafii, M.H. Ahmadi, W.M. Yan, M.A. Nazari, Int. J. Heat Mass Transf. 127 (2018) 203–208, https://doi.org/10.1016/j.ijheatmasstransfer.2018.06.108.
[14] S. Kianifard, M. Zamen, A.A. Nejad, J. Clean. Prod. 253 (2020), https://doi.org/10.1016/j.jclepro.2020.119972 119972.
[15] M. Yu et al., "Appl. Energy, vol. 256, no. April, p. 113929, 2019, doi: 10.1016/j.apenergy.2019.113929.
[16] A. Sharma, V. V. Tyagi, C. R. Chen, and D. Buddhi, "Renewable and Sustainable Energy Reviews, vol. 13, no. 2. pp. 318–345, Feb. 2009, doi: 10.1016/j.rser.2007.10.005.
[17] S.A. Nada, D.H. El-Nagar, Renew Energy 127 (2018) 630–641, https://doi.org/10.1016/j.renene.2018.05.010.
[18] A. Sá, R. Almeida, H. Sousa, J. Delgado, Adv. Mater. Sci. Eng. 2014 (2014), https://doi.org/10.1155/2014/582536.
[19] L. Tan, A. Date, G. Fernandes, B. Singh, S. Ganguly, Energy Procedia 110 (2017) 83–88, https://doi.org/10.1016/j.egypro.2017.03.110.
[20] R. Stropnik, U. Stritih, Renew. Energy 97 (2016) 671–679, https://doi.org/10.1016/j.renene.2016.06.011.
[21] R. M., L. S., R. S., A. H., and D. A., Sol. Energy, vol. 188, no. February, pp. 327–338, 2019, doi: 10.1016/j.solener.2019.05.067.
[22] Z. Li, T. Ma, J. Zhao, A. Song, Y. Cheng, Energy 178 (2019) 471–486, https://doi.org/10.1016/j.energy.2019.04.166.
[23] D. M. C. Shastry and U. C. Arunachala, J. Energy Storage, vol. 28, no. February, p. 101312, 2020, doi: 10.1016/j.est.2020.101312.
[24] R. Wang, J. Wang, and W. Yuan, Micromachines, vol. 10, no. 9, 2019, doi: 10.3390/mi10090620.
[25] P. Valeh-e-sheyda, M. Rahimi, E. Karimi, M. Asadi, ENERGY Convers. Manag. 69 (2013) 122–130, https://doi.org/10.1016/j.enconman.2013.01.029.
[26] M. Rahimi, E. Karimi, M. Asadi, P. Valeh-e-sheyda, Int. Commun. Heat Mass Transf. 43 (2013) 131–137, https://doi.org/10.1016/j.icheatmasstransfer.2013.02.002.
[27] M. Ali, H. M. Ali, W. Moazzam, and M. Babar Saeed, AIMS Energy, vol. 3, no. 4, pp. 699–710, 2015, doi: 10.3934/energy.2015.4.699.
[28] M. Benghanem, A.A. Al-Mashraqi, K.O. Daffallah, Renew. Energy 89 (2016) 51–59, https://doi.org/10.1016/j.renene.2015.12.011.
[29] H. Najafi, K.A. Woodbury, Sol. Energy 91 (2013) 152–160, https://doi.org/10.1016/j.solener.2013.01.026.
[30] M.D.S. Borkar, D.S.V. Prayagi, M.J. Gotmare, Int. J. Eng. Res. 3 (9) (2014) 536–539, https://doi.org/10.17950/ijer/v3s9/904.